

Bayesian Quantile Deep Echo State Networks for Nonlinear Time Series

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Abstract

Conditional quantiles are central to asymmetric decision losses, tail-risk assessment, and interval forecasts, but Bayesian quantile regression for nonlinear time series can be difficult when the temporal feature vector is high-dimensional. We develop the quantile deep echo state network (Q-DESN), a Bayesian quantile regression model conditional on fixed features generated by a deep echo state network. Conditional on a specified reservoir construction and feature map, posterior uncertainty is assigned to the regression coefficients, likelihood, and shrinkage parameters. Single-level fits use asymmetric Laplace or quantile-fixed generalized asymmetric Laplace working likelihoods with ridge or regularized-horseshoe priors. Posterior computation uses Markov chain Monte Carlo when computationally practical and a model-specific variational Bayes approximation for analyses requiring repeated fitting. For quantile grids, we compare independent level-wise regressions followed by monotone rearrangement with a joint quantile-vector regression that shrinks adjacent quantile-specific coefficient differences. Synthetic studies, a single-origin retrospective Global Flood Awareness System (GloFAS) streamflow case, and a retrospective PriceFM comparison identify settings where this fixed-feature Bayesian regression improves finite-grid quantile scores.

1 Introduction

Conditional quantiles are natural summaries for asymmetric decision losses, tail-risk assessment, and interval forecasts. If $\mathcal{F}_t$ is the information available at forecast origin t , the level- p quantile of a future response is

$$Q_{t,h}(p) = \inf\{u : \Pr(Y_{t+h} \leq u \mid \mathcal{F}_t) \geq p\}, \quad 0 < p < 1.$$

Quantile regression estimates this quantity directly (Koenker and Bassett, 1978; Koenker, 2005). Dynamic applications are more difficult when nonlinear temporal structure must be represented by a large number of correlated predictors and inference must be repeated over forecast origins or probability levels.

Echo state networks (ESNs) represent temporal dependence through recurrent nonlinear states whose reservoir weights are drawn once rather than estimated with the regression coefficients (Jaeger, 2001; Lukoševičius and Jaeger, 2009). Deep ESNs stack these reservoirs to obtain features at several time scales (Gallicchio et al., 2018). We use the resulting states as a fixed nonlinear design and place Bayesian quantile regression on the final linear predictor. This separation retains the computational economy of reservoir methods while providing regularized posterior inference for the conditional quantile.

The proposed quantile deep echo state network (Q-DESN) uses either the asymmetric Laplace (AL) likelihood or the quantile-fixed generalized asymmetric Laplace likelihood of Yan et al. (2025), abbreviated exAL. Both are working likelihoods: they define posterior inference for quantile-regression parameters without asserting the corresponding error law (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). The exAL family adds a shape parameter while preserving the chosen conditional quantile as its location. Ridge regularization supplies a dense baseline, and the regularized horseshoe (RHS) supplies adaptive global-local shrinkage with a finite slab (Carvalho et al., 2010; Piironen and Vehtari, 2017).

Several ESN methods already address probabilistic or quantile forecasting (McDermott and Wikle, 2019; Lv et al., 2016; Bonas et al., 2024). Q-DESN differs by treating a specified DESN realization as a fixed feature map and assigning a posterior to the quantile-specific regression, likelihood, and shrinkage parameters. Its linear dynamic comparators are the DQLM and exDQLM (Gonçalves et al., 2020; Barata et al., 2022), implemented in version 1.1.1 of the CRAN package `exdqmlm` (Barata et al., 2026).

For a grid of probability levels, we distinguish two inferential strategies. Independent regressions may be rearranged after fitting to obtain an ordered reported grid (Barlow and Brunk, 1972; Chernozhukov et al., 2010). A joint quantile-vector regression instead shares information during estimation by placing RHS priors on adjacent quantile-specific coefficient differences, adapting the Quantile-Varying Parameter construction of Kohns and Szendrei (2025). This prior can reduce crossings but does not enforce noncrossing; an ordered parameterization or explicit rearrangement is still required for exact monotonicity (Wu and Liu, 2009; Bondell et al., 2010; Yang and Tokdar, 2017).

The article contributes the fixed-feature Bayesian Q-DESN formulation, its independent and joint quantile-grid variants, and posterior computation by Markov chain Monte Carlo (MCMC) and a model-specific variational Bayes approximation. A single-quantile simulation compares Q-DESN with DQLM and exDQLM; a second simulation isolates joint versus independent estimation of a seven-level quantile grid. Retrospective GloFAS streamflow and PriceFM electricity-price examples assess independent Q-DESN regressions under their stated information sets. The supplement contains the complete posterior derivations, initialization and forecast details, diagnostics, and full numerical results.

2 Fixed-Feature Bayesian Quantile Models

2.1 Deep echo state network features

Let $\mathbf{u}_t = (y_{t-1}, \dots, y_{t-m}, \mathbf{z}_t^\top)^\top$ contain response lags and exogenous variables. For layer d , $\bar{\mathbf{W}}_d$ is a fixed sparse recurrent matrix scaled to target spectral radius ρ_d , $\mathbf{W}_d^{\text{in}}$ is a fixed input matrix, α_d is the leak rate, and $\mathbf{Q}_{d-1}$ reduces the preceding layer. With $\tilde{\mathbf{h}}_{t,d-1} = \mathbf{Q}_{d-1}\mathbf{h}_{t,d-1}$, the state recursion is

$$\begin{aligned} \mathbf{h}_{t,1} &= (1 - \alpha_1)\mathbf{h}_{t-1,1} + \alpha_1 f(\bar{\mathbf{W}}_1\mathbf{h}_{t-1,1} + \mathbf{W}_1^{\text{in}}\mathbf{u}_t), \\ \mathbf{h}_{t,d} &= (1 - \alpha_d)\mathbf{h}_{t-1,d} + \alpha_d f(\bar{\mathbf{W}}_d\mathbf{h}_{t-1,d} + \mathbf{W}_d^{\text{in}}\tilde{\mathbf{h}}_{t,d-1}), \quad d = 2, \dots, D, \\ \tilde{\mathbf{x}}_t &= [\mathbf{h}_{t,D}^\top, k(\tilde{\mathbf{h}}_{t,1})^\top, \dots, k(\tilde{\mathbf{h}}_{t,D-1})^\top, \mathbf{u}_t^\top]^\top, \quad \mathbf{x}_t = (1, \tilde{\mathbf{x}}_t^\top)^\top. \end{aligned} \quad (2.1)$$

The reservoir matrices, reducers, transformations, scaling quantities, and washout are determined from the fitting information and then held fixed. Posterior inference is therefore conditional on $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_T)^\top$. The supplement gives the random weight construction, dimensions, initialization, and washout conventions.

2.2 Single-level Q-DESN

At probability level p_0 , the marginal working model is

$$y_t \mid \mathbf{x}_t, \boldsymbol{\beta}, \sigma, \gamma \sim \text{exAL}_{p_0}(\mathbf{x}_t^\top \boldsymbol{\beta}, \sigma, \gamma), \quad t = 1, \dots, T, \quad (2.2)$$

where $\mathbf{x}_t^\top \boldsymbol{\beta}$ is the fitted conditional p_0 -quantile, $\sigma > 0$ is a scale, and $\gamma \in (L, U)$ controls shape. The quantile-fixing construction determines (L, U) and functions $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$. Computation uses

$$\begin{aligned} v_t \mid \sigma &\sim \text{Exp}(\text{rate} = 1/\sigma), & s_t &\sim \mathcal{N}^+(0, 1), \\ y_t \mid \cdot &\sim \mathcal{N}\left(\mathbf{x}_t^\top \boldsymbol{\beta} + \lambda(\gamma)\sigma s_t + A(\gamma)v_t, B(\gamma)\sigma v_t\right). \end{aligned} \quad (2.3)$$

Integrating out (s_t, v_t) recovers Eq. (2.2). The AL model is the nested case with its probability-specific constants and no s_t or γ block. Posterior intervals and forecast bands are thus model-based summaries conditional on the fixed feature map and working likelihood; their empirical coverage is assessed separately.

The intercept has a weak Gaussian prior. For the r non-intercept coefficients, ridge regularization uses $\boldsymbol{\beta}_{\text{slope}} \sim \mathcal{N}(\mathbf{0}_r, \kappa_\beta^2 \mathbf{I}_r)$. The adaptive alternative uses the Nishimura–Suchard product form of the regularized horseshoe (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). Conditional on global, local, and slab scales (τ, λ_j, ζ) ,

$$\beta_j \mid \tau, \lambda_j, \zeta \sim \mathcal{N}(0, V_j), \quad V_j = \left(\zeta^{-2} + \tau^{-2}\lambda_j^{-2}\right)^{-1}, \quad (2.4)$$

with $\lambda_j \sim C^+(0, 1)$ and $\tau \sim C^+(0, \tau_0)$. The product density and all scale-dependent normalizing factors are stated in the supplement.

Reference calibration of the global scale. The hyperprior scale τ_0 controls overall shrinkage.

With standardized predictors, N_{fit} fitting observations, an elicited reference $m_0 \in (0, r)$ for the number of coefficients expected to escape strong shrinkage, and residual scale σ_{ref} , the Gaussian reference calibration of Piironen and Vehtari (2017) is

$$\tau_{0,\text{ref}} = \frac{m_0}{r - m_0} \frac{\sigma_{\text{ref}}}{\sqrt{N_{\text{fit}}}}. \quad (2.5)$$

We recommend this common reference when comparing the Gaussian, AL, and exAL models on the same standardized design; it fixes the coefficient-prior scale while each likelihood retains its own posterior. Likelihood-information calibrations and separate choices for joint-model coefficient blocks are given in the supplement.

2.3 Joint quantile-vector regression

For $0 < p_1 < \dots < p_Q < 1$, the joint model uses

$$Q_{p_q}(y_t \mid \mathcal{F}_t) = \alpha_q + \tilde{\mathbf{x}}_t^\top \boldsymbol{\beta}_q, \quad q = 1, \dots, Q, \quad (2.6)$$

and multiplies the Q quantile-specific AL or exAL contributions. This is an untempered composite working likelihood for repeated use of the response series. Let $\boldsymbol{\beta}_{\text{base}}$ be a baseline slope and define

$$\boldsymbol{\Delta}_1 = \boldsymbol{\beta}_1 - \boldsymbol{\beta}_{\text{base}}, \quad \boldsymbol{\Delta}_q = \boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}, \quad q = 2, \dots, Q.$$

Separate RHS hierarchies are assigned to the baseline and to each $\boldsymbol{\Delta}_q$. This distinguishes effects shared across levels from departures between neighboring levels. For $Q = 1$, the baseline block is omitted and the RHS prior is placed directly on $\boldsymbol{\Delta}_1 = \boldsymbol{\beta}_1$.

Adjacent-difference shrinkage favors similar neighboring slopes, but the gap

$$Q_{p_q}(y_t \mid \mathcal{F}_t) - Q_{p_{q-1}}(y_t \mid \mathcal{F}_t) = (\alpha_q - \alpha_{q-1}) + \tilde{\mathbf{x}}_t^\top (\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1})$$

need not be positive. We consequently report crossings before rearrangement and apply an explicit monotone reporting rule when ordered output is required. The stacked Gaussian representation, block precision, and full joint posterior are given in the supplement.

3 Posterior Computation

The auxiliary representation in Eq. (2.3) yields Gaussian coefficient updates, generalized inverse-Gaussian mixture updates, and inverse-gamma RHS scale updates. For exAL, the scale-asymmetry block remains nonconjugate. MCMC updates it on transformed coordinates; the independent simulation uses a partially collapsed transition that integrates out the scale while updating γ and then redraws the scale from its full conditional. The joint model applies the corresponding update separately at each probability level. Convergence assessment includes rank-normalized $\hat{R}$, effective

sample sizes, chain comparisons of reported functionals, and quantile-crossing diagnostics.

For analyses requiring many repeated fits, we use a mean-field variational approximation to the same augmented posterior. Gaussian, GIG, truncated-Normal, and inverse-gamma factors retain closed-form coordinate updates. The nonconjugate exAL scale–asymmetry factor uses the Laplace–Delta approximation of Wang and Blei (2013); we denote this model-specific procedure by VB–LD. The evidence lower bound, full conditional distributions, algorithms, and joint sparse Gaussian calculation are collected in the supplement. Variational intervals are approximate and are identified separately from MCMC posterior intervals.

4 Forecasting, Scoring, and Model Selection

At forecast origin T , posterior or variational draws are propagated through the fixed reservoir recursion using covariates available at the origin or specified by the analysis. If future response lags are required, they are generated under the stated recursion or supplied by a stated scenario. For a single-level draw, the fitted quantile location at horizon h is $\mu_{p,T+h}^{(b)} = \mathbf{x}_{T+h}^{(b)\top} \boldsymbol{\beta}_p^{(b)}$. This is a draw of the fitted conditional quantile. Sampling the AL or exAL working likelihood instead produces a response trajectory; the two summaries are not interchangeable. The supplement gives the complete recursive constructions.

For independent fits on $\mathcal{P}_K = \{p_1, \dots, p_K\}$, with $K \geq 2$, the raw estimates may cross. We write $q_{T,h,k}^*$ for the reported value after the monotone transformation. The reported analyses use equal-weight isotonic projection,

$$\mathbf{q}_{T,h}^* = \arg \min_{z_1 \leq \dots \leq z_K} \sum_{k=1}^K (\hat{q}_{T,h,k} - z_k)^2, \quad (4.1)$$

with the study-specific choice of draw-level or summary-level projection stated with the results. In analyses based on summary-level projection, the posterior-mean quantile curve or variational posterior-mean quantile estimates are transformed before scoring. Linear interpolation defines the quantile curve only over $[p_1, p_K]$; a complete predictive distribution also requires a tail rule.

Let the check loss be

$$\rho_p(e) = e\{p - \mathbf{1}(e < 0)\}. \quad (4.2)$$

For a predictive distribution F with finite first moment, the CRPS has the integrated check-loss representation (Gneiting and Raftery, 2007)

$$\text{CRPS}(F, y) = 2 \int_0^1 \rho_p\{y - F^{-1}(p)\} dp. \quad (4.3)$$

For a finite evaluation grid $\mathcal{P}_K = \{p_1, \dots, p_K\}$, we report the trapezoidal approximation over its

covered range,

$$\text{aCRPS}_{\mathcal{P}_K}(y; \mathbf{q}^*) = 2 \sum_{k=1}^K \omega_k \rho_{p_k}(y - q_k^*), \quad \omega_1 = \frac{p_2 - p_1}{2}, \quad \omega_k = \frac{p_{k+1} - p_{k-1}}{2}, \quad \omega_K = \frac{p_K - p_{K-1}}{2}. \quad (4.4)$$

Here the middle expression applies for $2 \leq k \leq K-1$, and $\sum_{k=1}^K \omega_k = p_K - p_1$. Thus aCRPS retains the scale of integrated check loss over the fitted range rather than representing a full-distribution CRPS.

In the joint simulation the true conditional response distribution is known. For held-out origin–horizon pairs $r \in \mathcal{G}$, posterior draw b , and rearranged quantiles $q_{r,k}^{*,(b)}$, we also use

$$S_{\text{DGP}}^{(b)} = \frac{2}{|\mathcal{G}|} \sum_{r \in \mathcal{G}} \sum_{k=1}^K \omega_k \mathbb{E}_{Y_r \sim F_{0,r}} [\rho_{p_k}\{Y_r - q_{r,k}^{*,(b)}\}]. \quad (4.5)$$

This finite-grid score averages over the known conditional response distribution at the fixed held-out design points. Its interval excludes repeated-simulation and model-selection uncertainty; the supplement records the coupling used for independently fitted levels.

For a statistical model $\mathcal{M}$, reservoir candidates $\mathcal{C} = \{\mathcal{R}_1, \dots, \mathcal{R}_J\}$ are compared by a stated validation criterion,

$$\hat{\mathcal{R}}_{\mathcal{M}} \in \arg \min_{\mathcal{R}_j \in \mathcal{C}} S_{\text{val}}(\mathcal{R}_j; \mathcal{M}), \quad (4.6)$$

and the selected specification is fixed before held-out evaluation. The response convention, information set, forecast origins and horizons, transformations, feature ordering, and score remain common within a comparison. State checks cover finite values, activation saturation, near-constant or highly dependent columns, effective rank, conditioning, and variation across reservoir realizations. These checks are needed because spectral-radius scaling alone does not guarantee suitable driven dynamics (Jaeger et al., 2007; Yildiz et al., 2012).

The joint simulation and both applications use sequential initialization centered at the median. A Gaussian ridge fit initializes Gaussian RHS–VB; compatible estimates then initialize the median AL regression. AL fits proceed from the median toward each tail, each exAL fit starts from the matching AL level, and the complete independent grids initialize their corresponding joint models. A VB fit initializes MCMC only for the same statistical model. These steps alter starting values, not the destination likelihood, prior, posterior, or independence of level-wise fits. The independent validation study uses a different procedure. The complete sequence, compatibility conditions, and diagram are provided in the supplement.

5 Simulation Studies

The simulation study addresses two complementary questions. The first asks whether a single-quantile Q–DESN regression can recover and forecast a known dynamic quantile path under a 500-observation benchmark following the family-based comparison strategy of Yan et al. (2025). The

target quantile level is fixed in advance, the data-generating mechanism is known, and competing models are compared under matched error families, fitting-sample sizes, and quantile levels. The second asks whether a joint multi-quantile regression can recover several known conditional quantile paths under posterior simulation and reduce adjacent-level crossings before monotone rearrangement. The supplement reports the corresponding VB analyses and additional forecast comparisons.

In contrast to the linear-regression setting of Yan et al. (2025), the true conditional quantile paths, referred to as oracle paths in this simulation, are dynamic. Q-DESN represents those paths through high-dimensional deterministic reservoir features. The design therefore separates two questions: how accurately a fitted model recovers the known conditional quantile path, and how well it forecasts held-out dynamic blocks. We write p for the single target level in the first study and τ for levels in the multi-quantile grid.

The main-text tables report MCMC summaries under fixed evaluation designs: data-generating mechanisms, rolling-origin definitions, fitting samples, held-out periods, scoring rules, and model classes are specified before the comparisons are formed. The supplement gives corresponding approximate VB summaries and additional diagnostic summaries for the same simulation settings. The single-quantile and multi-quantile studies use separate fixed designs. Their results address distinct comparisons and provide complementary evidence.

5.1 Single-Quantile Dynamic Data-Generating Processes

For each error family f and target quantile level $p \in \{0.05, 0.25, 0.50\}$, the synthetic series is generated as

$$y_{t,p,f} = \mu_{t,f} + \varepsilon_{t,p,f}, \quad \varepsilon_{t,p,f} = e_{t,f} - H_f^{-1}(p),$$

where H_f is the distribution function of the raw error $e_{t,f}$. The shift by $H_f^{-1}(p)$ makes the p -quantile of $\varepsilon_{t,p,f}$ equal to zero. Hence

$$Q_p(y_{t,p,f} \mid \boldsymbol{\theta}_{t,f}) = \mu_{t,f}.$$

The true conditional quantile path is generated by a six-dimensional dynamic linear state with local level, local slope, and two harmonic pairs:

$$\mu_{t,f} = \mathbf{F}^\top \boldsymbol{\theta}_{t,f}, \quad \mathbf{F} = (1, 0, 1, 0, 1, 0)^\top,$$

with local linear trend and two sine–cosine harmonic pairs. The seasonal period is 90, the harmonics are 1 and 2, and the state innovation covariance $\mathbf{W}_\theta$ has diagonal square roots

$$\sqrt{\text{diag}(\mathbf{W}_\theta)} = (0.005, 0.00002, 0.004, 0.004, 0.003, 0.003).$$

The initial covariance is $\mathbf{C}_0 = 0.01\mathbf{I}_6$, and each simulated trajectory is initialized deterministically at the family-specific state in Table 1. For a fixed family, the same latent path $\{\mu_{t,f}\}$ is used across the three values of p ; only the centering constant $H_f^{-1}(p)$ changes.

Family	Raw error distribution H_f	Initial level	Initial slope	Seasonal settings
Gaussian	$\mathcal{N}(0, 10^2)$	40	0.012	harmonics (24, 0.35), (8, -0.80)
Laplace	Laplace with scale 10	35	0.011	harmonics (28, -0.15), (10, 0.75)
Gaussian mixture	$0.1\mathcal{N}(0, 0.5^2) + 0.9\mathcal{N}(1, 15^2)$	45	0.014	harmonics (32, 0.85), (12, -1.25)

Table 1: Single-quantile dynamic data-generating mechanisms. Raw errors are shifted by $H_f^{-1}(p)$ so that the p -quantile of the observation error is zero; the same latent dynamic quantile path is then shared across target levels within each family. Seasonal entries report amplitude and phase for the two harmonic pairs in the initial state.

5.2 Single-Quantile Fit and Forecast Evaluation

Each synthetic series is first run for $T_{\text{warm}} = 2000$ warmup observations, which are discarded before analysis. The next $T_{\text{main}} = 10000$ observations form the retained series. The first 9000 support model specification and fitting, and the final 1000 are held out for forecast evaluation. The reported single-quantile comparison uses $T_{\text{fit}} = 500$, namely observations 8501–9000 immediately preceding the fitting origin. Forecast scores are computed from rolling origins within the held-out period; origins are spaced 30 observations apart, and each origin is scored for horizons up to 30 steps ahead. The same evaluation design is used for all model specifications and score calculations.

5.3 Single-Quantile Competing Methods

For each family and target quantile level in the single-quantile simulation comparison, the reported Q-DESN rows vary the working likelihood and inference method while using regularized-horseshoe (RHS) coefficient shrinkage. Entries labeled Q-DESN AL-RHS use the AL working likelihood, whereas entries labeled Q-DESN exAL-RHS use the exAL working likelihood. The main comparison focuses on regularized-horseshoe fits. For each displayed metric, the reservoir specification and realization were selected before held-out evaluation and then held fixed. Dynamic quantile linear model (DQLM) and extended dynamic quantile linear model (exDQLM) baselines are fit to the same family, quantile level, fitting-sample size, and forecast-evaluation design using version 1.1.1 of the CRAN package `exdqml` (Barata et al., 2026). The exDQLM calculations use its partially collapsed MCMC update and structured $q(\gamma)q(\sigma \mid \gamma)$ variational approximation for the scale-asymmetry block. For rolling-origin MCMC evaluation, the initial parameter posterior is retained while the filtering distribution is updated sequentially using the responses available at each origin. In the exDQLM analysis, the observation-error mean and variance used in this update are averaged over paired retained draws of the scale and asymmetry parameters. The updated state is then propagated over the requested forecast horizons.

The primary comparison is a fixed-design benchmark against structured dynamic quantile linear models under linear dynamic mechanisms that favor those baselines. Repeated data-generating

realizations, broader reservoir-parameter comparisons, and non-Bayesian ESN regression baselines remain topics for future sensitivity analyses.

5.4 Single-Quantile Criteria for Comparison

Let $q_{p,t}^{(b)}$ be retained draw b of the conditional p -quantile at fitting index t , and let $q_{p,r}^{(b)}$ be the corresponding forecast draw for rolling-origin pair r . Write $\mathcal{I}$ for the 500 fitting indices and $\mathcal{G}$ for the 1000 scored origin–horizon pairs. Because the oracle path μ is known, the draw-wise criteria are

$$R_{\text{fit}}^{(b)} = \left\{ \frac{1}{|\mathcal{I}|} \sum_{t \in \mathcal{I}} (q_{p,t}^{(b)} - \mu_t)^2 \right\}^{1/2}, \quad (5.1)$$

$$A_{\text{fore}}^{(b)} = \frac{1}{|\mathcal{G}|} \sum_{r \in \mathcal{G}} |q_{p,r}^{(b)} - \mu_r|, \quad (5.2)$$

$$C_{\text{fore}}^{(b)} = \frac{1}{|\mathcal{G}|} \sum_{r \in \mathcal{G}} \rho_p(y_r - q_{p,r}^{(b)}), \quad (5.3)$$

where ρ_p is the check loss in Eq. (4.2), and μ_r and y_r denote the oracle quantile and realized outcome for pair r . Thus fit RMSE and forecast MAE measure recovery of the oracle quantile path, whereas forecast check loss is a proper score against realized observations.

For each criterion $M^{(b)}$, the tables report the posterior mean $B^{-1} \sum_{b=1}^B M^{(b)}$ and the empirical 0.025 and 0.975 posterior quantiles. The VB panels use draws from the variational approximation and are therefore approximate. Because the criteria are nonlinear functions of the quantile path, these posterior means generally differ from applying a criterion once to a posterior point path.

For each model, error family, quantile level, and criterion, the reservoir specification was selected in a separate calibration analysis before the posterior metric distribution was evaluated. Consequently, different criteria within a model may correspond to different reservoir specifications. The intervals condition on the selected specification, simulated series, rolling-origin design, and reservoir realization; repeated-simulation, reservoir-selection, and model-selection uncertainty are excluded.

5.5 Single-Quantile Fit and Forecast Results

Figures 1 and 2 summarize the MCMC posterior distributions of the two forecast criteria. Among the 27 family–quantile–criterion comparisons, Q–DESN exAL–RHS has the lowest posterior mean in 10 and Q–DESN AL–RHS in 8, compared with 4 for DQLM and 5 for exDQLM. The equal-tailed intervals of the two lowest means overlap in every comparison, and five of the 108 displayed MCMC summaries carry diagnostic cautions.

The displayed bands are posterior distributions of the draw-wise aggregate criteria. For Q–DESN and exQ–DESN, each metric draw preserves a common posterior draw of the quantile-regression coefficients across forecast origins and horizons. For DQLM and exDQLM, the origin- and horizon-specific posterior quantile marginals are combined using a pre-specified product coupling. The

resulting intervals quantify model-specific posterior uncertainty for the reported criteria, conditional on the simulated series, selected specification, and evaluation design.

Posterior means were numerically unchanged across coupling assumptions, whereas interval widths varied; we therefore report the product-coupling intervals with this sensitivity analysis.

All preprocessing uses fitting information only. Complete MCMC and approximate VB values, fit-recovery results, warning details, and a five-chain sensitivity analysis are reported in the supplement.

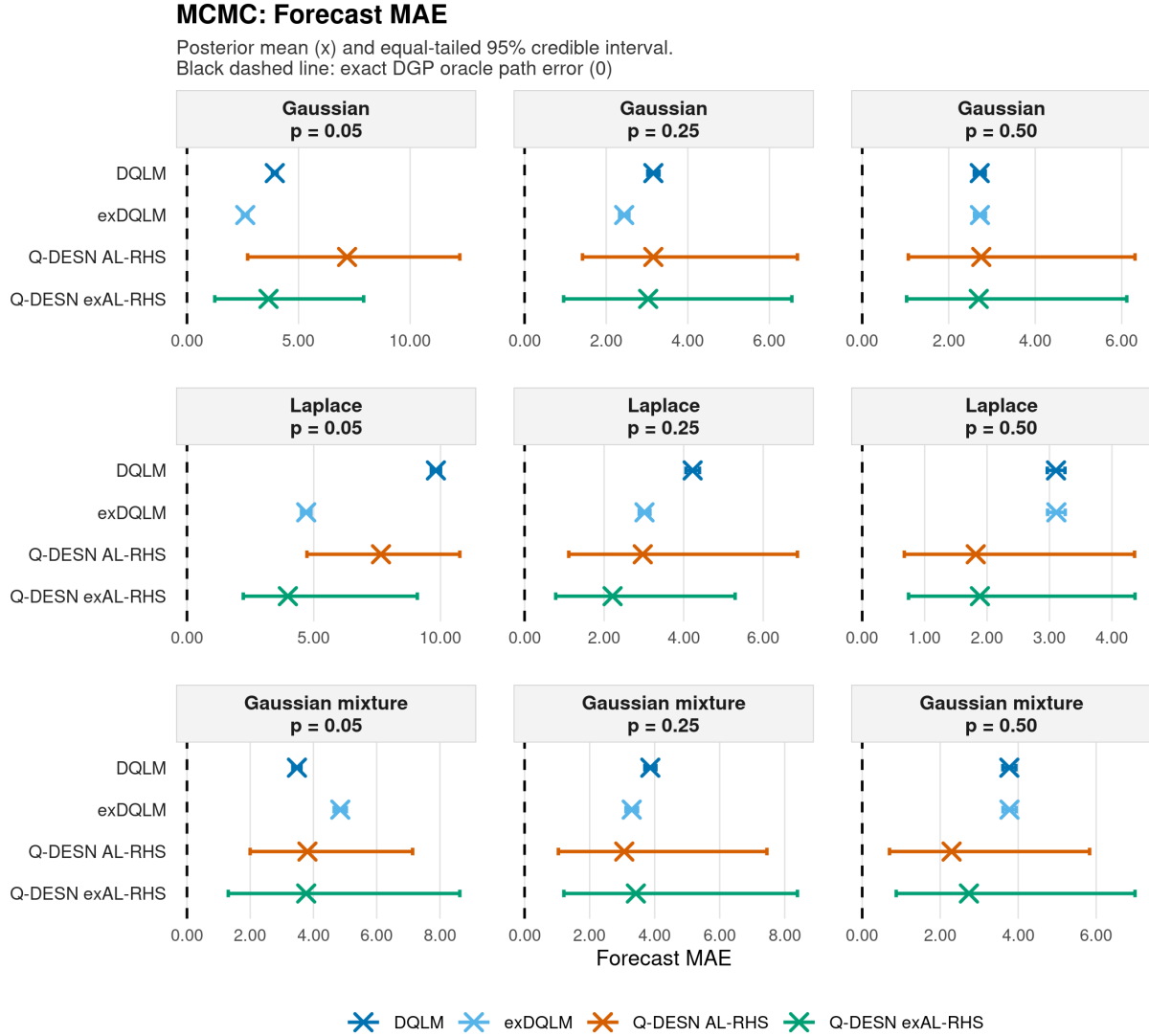


Figure 1: MCMC posterior uncertainty for forecast MAE in the single-quantile simulation. Horizontal segments show equal-tailed 95% posterior intervals and crosses show posterior means. Each panel has its own horizontal scale; lower is better. The dashed line marks the exact DGP value of zero for quantile-path error. Intervals condition on the simulated series, evaluation design, and case-specific selected specification.

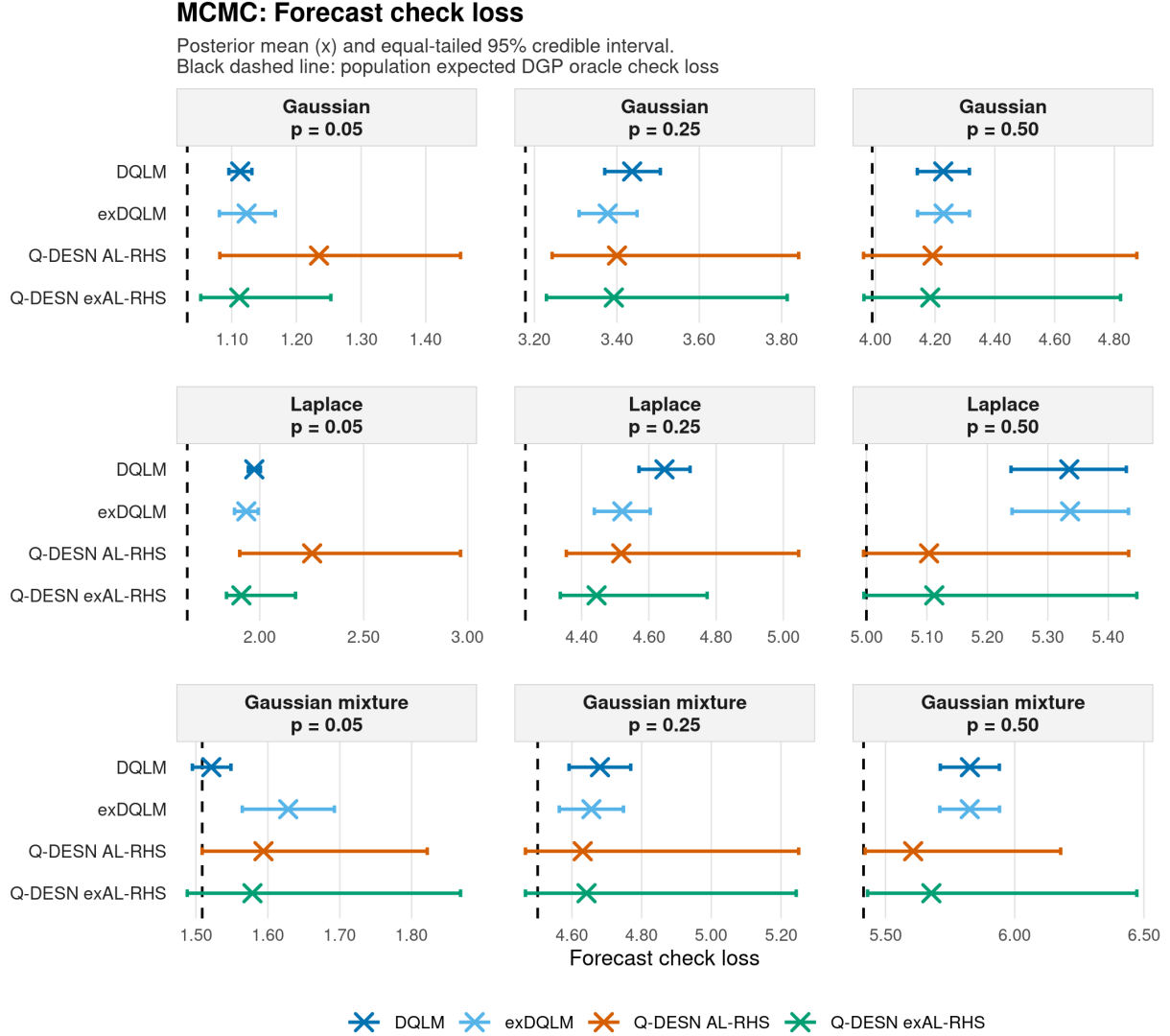


Figure 2: MCMC posterior uncertainty for forecast check loss in the single-quantile simulation. Horizontal segments show equal-tailed 95% posterior intervals and crosses show posterior means. Each panel has its own horizontal scale; lower is better. The dashed line marks population expected check loss at the true conditional quantile. Conditional finite-sample intervals may cross this reference.

5.6 Joint Multi-Quantile Study with a Shared Feature Design

The second simulation study compares joint and independent estimation of the seven levels

$$\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$$

under the AL and exAL working likelihoods. Eight mechanisms span Gaussian, Laplace, Gaussian-mixture, asymmetric-tail, persistent-heavy-tail, Student- t , regime-shift, and nonlinear dynamics. For each mechanism, one DESN and RHS specification is selected using separate source realizations and then shared by all four model classes. The 500-observation evaluation realization is not used for this selection. Thus differences among the 32 comparisons are not confounded by model-specific feature maps or prior scales.

The evaluation fits use sequential initialization centered at the median. A Gaussian ridge fit initializes Gaussian RHS-VB; the resulting compatible estimates initialize the median AL regression; and the remaining independent AL fits proceed toward the two tails. Each exAL fit is initialized from the AL fit at the same level, the complete independent grids initialize the corresponding joint fits, and each completed VB fit initializes MCMC for the same model. These transfers determine starting values only. The supplement gives the complete calculation and chain allocation.

Held-out evaluation is sequentially conditional: regression coefficients and reservoir weights remain fixed, while newly observed response lags enter later forecast origins. Each comparison contains the same 990 origin-horizon rows. The primary criterion is S_{DGP} in Eq. (4.5), computed for each retained MCMC draw after monotone rearrangement. Independent quantile draws are combined by the pre-specified chain-balanced product coupling described in the supplement.

Figure 3 shows the complete posterior score comparison. Independent exQDESN has the smallest numerical posterior mean in seven mechanisms; joint exQDESN is smallest for the Laplace mechanism. The marginal intervals for the numerical winner and runner-up overlap in every mechanism. Of the 16 paired joint-minus-independent contrasts, five intervals favor independent estimation and eleven include zero; none favors joint estimation. The evidence therefore does not support a general forecast advantage from joint estimation under this shared design.

Before rearrangement, the posterior-mean forecast grids contain 1,426 crossings for joint Q-DESN, 4,705 for independent Q-DESN, none for joint exQDESN, and 281 for independent exQDESN, summed over the eight mechanisms. Rearrangement produces ordered grids in all 32 comparisons. Posterior-score diagnostics meet the pre-specified criteria in 21 comparisons and warrant review in 11; all 16 exAL scale-asymmetry assessments remain at review level. Exact scores, fit recovery, crossing rates, and computational diagnostics are reported in the supplement. A separate 50-replicate variational analysis is retained as a sensitivity study.

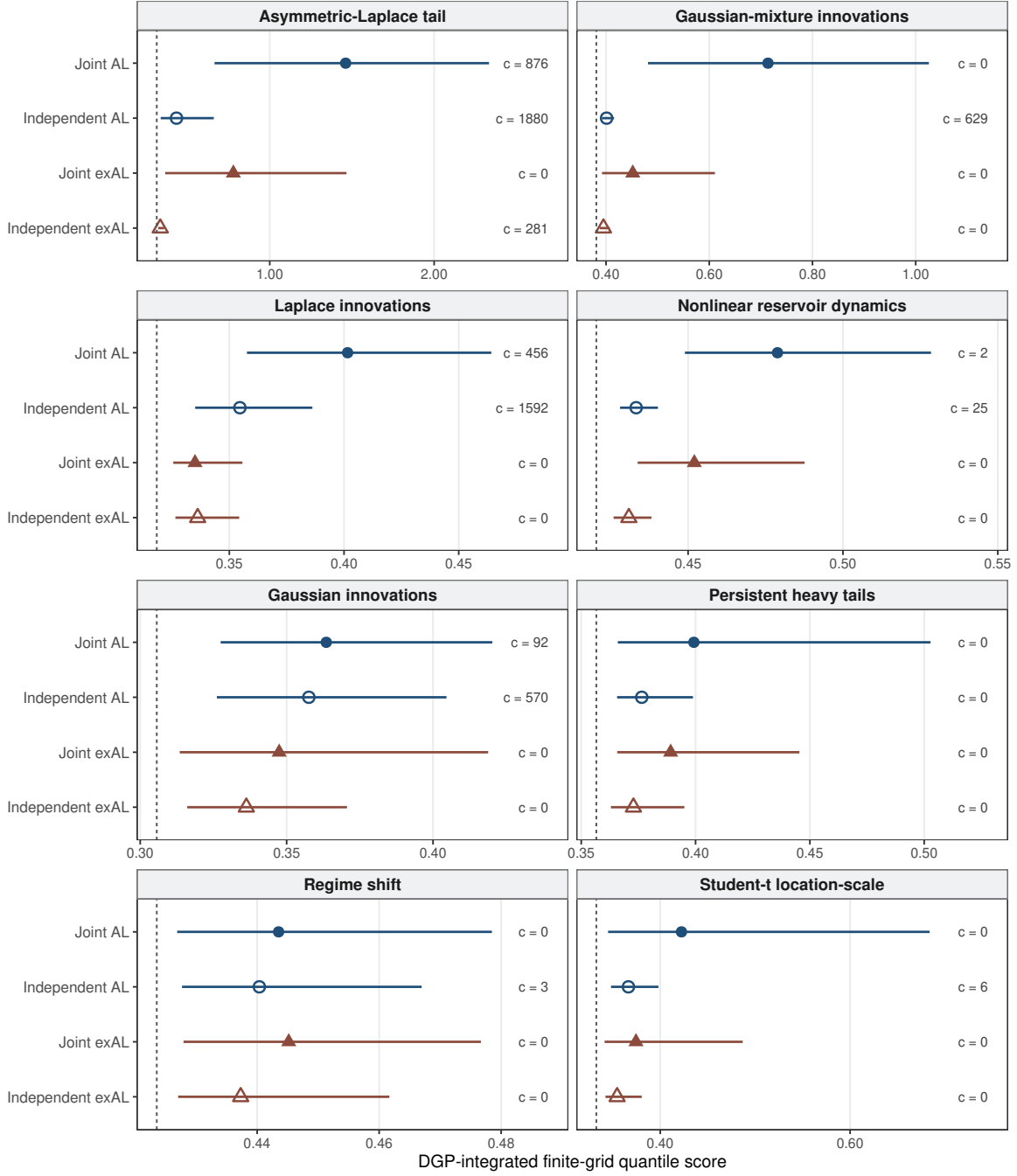


Figure 3: Posterior DGP-integrated finite-grid quantile scores under the shared-backbone design. Points are posterior means and horizontal bars are equal-tailed 95% intervals; lower is better. Dashed lines mark the oracle finite-grid score. Filled symbols denote joint fits and open symbols independent fits; blue denotes AL and red exAL. Labels c give raw crossings in the posterior-mean forecast grid; all counts after rearrangement are zero.

6 Application: GloFAS Retrospective Streamflow Quantile Forecasting

Retrospective case study. We consider a medium-range streamflow forecast from the Global Flood Awareness System (GloFAS), which is jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts and operated within the Copernicus Emergency Management Service (Alfieri et al., 2013; Copernicus Emergency Management Service, 2026). GloFAS provides ensemble river-discharge forecasts for flood early warning and verification (Harrigan et al., 2023). Q-DESN supplies a statistical quantile correction relative to a United States Geological Survey (USGS) gauge record at one forecast origin.

6.1 Data and Forecast-Origin Design

Let T be the forecast origin, y_t the transformed USGS record through T , g_t^{ret} the retrospective GloFAS series, and $g_{T+h,m}^{\text{ens}}$ issued ensemble member m at horizon h . The held-out y_{T+h} values are used only for evaluation. The analysis uses origin 2022-12-25 and scores seven probability levels over 28 target dates. Its covariate specification combines observed history and Global Ensemble Forecast System (GEFS) forecasts with realized-future contributions to precipitation and soil-moisture covariates. This blended information set makes the analysis retrospective; it does not represent a real-time forecast based only on information available at T .

6.2 Application Model

Persistence-anchored discrepancy. Two fixed Q-DESN feature maps represent the reference history and the retrospective discrepancy $g_t^{\text{ret}} - y_t$. For $t > T$, the persistence-plus-adjustment specification anchors the future discrepancy feature at its last historical value; subsequent discrepancy features describe adjustments around that persistence-anchored discrepancy baseline. For probability level p_0 ,

$$q_{p_0,t}^{\text{ref}} = \mathbf{x}_t^{\text{ref}\top} \boldsymbol{\beta}_{p_0}, \quad d_{p_0,t}^{\text{glo}} = \mathbf{x}_t^{\text{disc}\top} \boldsymbol{\alpha}_{p_0}, \quad q_{p_0,t}^{\text{glo}} = q_{p_0,t}^{\text{ref}} + d_{p_0,t}^{\text{glo}}.$$

The reference-process quantile $q_{p_0,t}^{\text{ref}}$ is the scored quantity, and $d_{p_0,t}^{\text{glo}}$ is the quantile-specific GloFAS discrepancy. The reported analysis fits the seven levels 0.05, 0.15, 0.35, 0.50, 0.65, 0.80, and 0.95 independently with AL working likelihoods, separate RHS hierarchies for the two coefficient blocks, and VB. Issued ensemble members are treated as conditionally independent and exchangeable given the forecast-system quantile path, a working-likelihood approximation because the members arise from perturbed initial conditions and model perturbations (European Centre for Medium-Range Weather Forecasts, 2026). Variational posterior-mean quantile estimates are monotonically rearranged on the fitted grid. The supplement gives the source-specific likelihood, the fixed reservoir specifications, and the more general exAL and latent-future-path formulations.

6.3 Retrospective Application Results

The reported fit was evaluated over 12,495 observed dates through the forecast origin. After monotone rearrangement, its observational-window aCRPS was 0.0445, and empirical 90% interval coverage was 0.906. These fitting-period scores were used for model selection, with the forecast period reserved for evaluation. The fitted $p_0 = 0.95$ quantile occasionally takes extreme values in the observational period, so upper-tail forecasts are interpreted cautiously.

Forecast-period scores. Table 2 shows lower mean check loss, interval score, and aCRPS for Q-DESN than for raw GloFAS at this origin. Empirical coverage of the nominal 90% interval is 0.357 for Q-DESN and 0.000 for raw GloFAS. The improvement from zero coverage does not resolve the substantial Q-DESN undercoverage. Multi-origin evaluation using only covariates available at each origin is needed to assess real-time performance and calibration.

Table 2: Forecast scoring for the GloFAS retrospective analysis. Scores are computed on the transformed streamflow scale for the forecast origin 2022-12-25. Lower check loss, interval score, and aCRPS are better; aCRPS is computed from Eq. (4.4) at seven common evaluation levels. Mean coverage reports empirical coverage for the nominal 90% interval and is interpreted descriptively. The final column gives the relative reduction in mean check loss.

Model	Horizons	Check	Interval	aCRPS	Coverage	Check reduction
Q-DESN correction	28	0.5654	10.8032	1.1319	0.357	26.0%
Raw GloFAS	28	0.7639	25.2028	1.4424	0.000	Reference

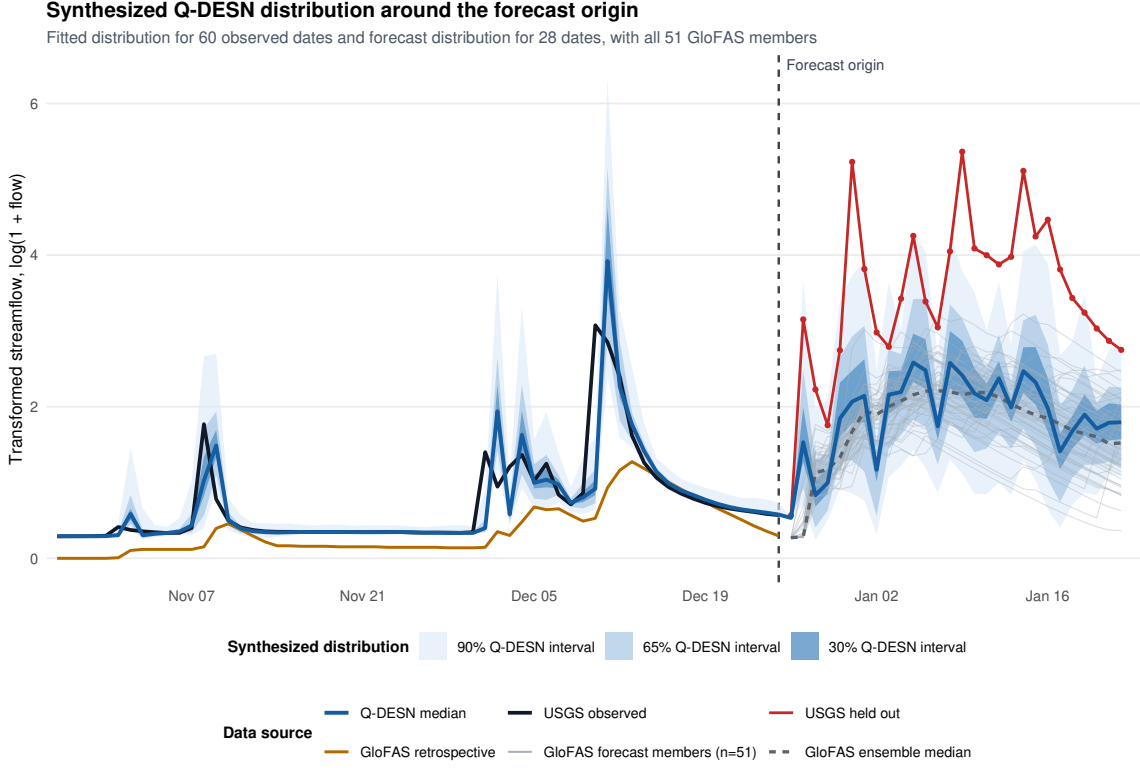


Figure 4: Q-DESN quantile intervals for the GloFAS retrospective analysis over the final 60 observed dates and the following 28-day forecast period. The nested blue bands are the 90%, 65%, and 30% quantile intervals, and the blue path is the median. The vertical dashed line marks the forecast origin. Black and red paths denote observed and held-out USGS streamflow, respectively; orange denotes retrospective GloFAS, and thin gray paths show all 51 issued GloFAS ensemble members, with their median dashed. Forecast scores use the held-out measurements in Table 2.

7 Application: PriceFM Retrospective Electricity-Price Comparison

The second application compares retrospective Q-DESN day-ahead electricity-price forecasts with version 4 of PriceFM (Yu et al., 2026). The release covers European bidding regions at 15-minute resolution from 1 January 2022 through 1 January 2026 and includes price, load, solar, and wind forecasts. Average quantile loss (AQL) is evaluated on the PriceFM probability grid at matched held-out timestamps for 114 region-fold comparisons.

7.1 Comparison Design

The comparison contains 114 region-fold evaluations, spanning 38 European bidding regions and 3 rolling folds. Each evaluation is scored on the fold-aligned held-out timestamps and probability levels 0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90. Throughout this section, Δ AQL denotes Q-DESN

AQL minus PriceFM AQL. Negative values indicate lower Q-DESN AQL; positive values indicate lower PriceFM AQL.

Retrospective covariates. We compare Q-DESN forecasts with the released PriceFM forecasts at matched region-fold timestamps. Across the comparison, 58 evaluations use own-region predictors and 56 evaluations also include neighboring-region summaries. Both predictor sets use retrospectively observed own-region load, solar, and wind lead covariates in the comparison; evaluations with neighborhood summaries add retrospectively observed neighboring-region lead summaries. The score comparison is therefore retrospective. A PriceFM study of the joint quantile-vector RHS prior would require the same fold-aligned design with that joint regression fitted directly.

7.2 Results by Region, Fold, and Forecast Lead

Across the aligned benchmark, Q-DESN has lower AQL in 66 of 114 region-fold evaluations (57.9%). Relative to its own AQL, PriceFM has lower AQL by at most 5% in 12 comparisons and by more than 5% in 36 comparisons. The mean Q-DESN AQL is 6.824, compared with mean PriceFM AQL 7.039, giving mean Δ AQL -0.215 and median Δ AQL -0.083. The differences vary across region-fold comparisons. Within 56 region-fold comparisons, AL and exAL specifications were chosen using validation AQL and then evaluated on held-out observations. The validation-selected specifications averaged test AQL 6.704, compared with 6.608 for the reference Q-DESN specifications. The reported aggregate uses the selected specification in 12 cases in which its held-out AQL was lower than both the reference Q-DESN and matched PriceFM values. Because held-out performance determined these 12 substitutions, the resulting aggregate is descriptive; complete case-level results are reported in the supplement.

Across the 114 region-fold comparisons, the unweighted mean AQCR is 2.58% for Q-DESN and 0.00% for PriceFM. Mean MAE is 16.722 for Q-DESN and 17.274 for PriceFM; mean RMSE is 25.449 and 26.336, respectively. Relative to PriceFM, Q-DESN has 3.1%, 3.2%, and 3.4% lower mean AQL, MAE, and RMSE, respectively. Table 3 summarizes the AQL results overall, by fold, and by forecast-lead block.

Mean Q-DESN AQL is lower in folds 1 and 2 and nearly equal in fold 3. Evaluations with neighborhood summaries have a more negative mean Δ AQL than own-region-only evaluations. Forecast-lead-specific results are available for 72 of the 114 comparisons: PriceFM is favored in the first six-hour block, whereas the later blocks favor Q-DESN on average but remain heterogeneous across regions and folds. The differing predictor sets, case-specific specifications, and incomplete lead-specific panel make these comparisons descriptive.

In this retrospective comparison, Q-DESN has lower aggregate AQL, MAE, and RMSE than the matched PriceFM forecasts, with substantial variation across folds, regions, and forecast horizons. Paired uncertainty analyses and direct evaluation of the joint quantile model remain topics for future study.

Table 3: Matched retrospective comparison of Q-DESN and PriceFM. Panel A reports unweighted summaries across region-fold comparisons, overall and by fold. The two PriceFM columns distinguish $0 < \Delta \text{AQL} / \text{AQL}_{\text{PriceFM}} \leq 0.05$ from relative differences greater than 0.05. Panel B reports paired AQL differences for the 72 region-fold comparisons with forecast-lead-specific results. Its means and medians are unweighted summaries of the 72 paired differences in each forecast-lead block. At 15-minute resolution, the four successive 24-step blocks each span six hours. Here ΔAQL is Q-DESN minus PriceFM, so negative values favor Q-DESN. AQL is in EUR/MWh, and all scores use matched held-out timestamps and the same seven probability levels.

Panel A: Overall and fold-specific results

Comparison set	n	Q-DESN lower	PriceFM lower by $\leq 5\%$	PriceFM lower by $> 5\%$	Mean Q-DESN AQL	Mean PriceFM AQL	Mean Δ	Median Δ
Overall	114	66	12	36	6.824	7.039	-0.215	-0.083
Fold 1	38	22	6	10	7.044	7.297	-0.253	-0.095
Fold 2	38	30	2	6	6.429	6.826	-0.397	-0.295
Fold 3	38	14	4	20	6.998	6.993	0.005	0.281

Panel B: Results by forecast-lead block

Forecast-lead block	n	Q-DESN lower	Mean Δ	Median Δ
1-24	72	22	0.415	0.351
25-48	72	45	-0.684	-0.382
49-72	72	34	-0.198	0.032
73-96	72	42	-0.155	-0.114

8 Discussion

Q-DESN treats the fixed DESN reservoir as a nonlinear feature map and places Bayesian quantile inference on the regression coefficients. Posterior simulation and variational inference concern the regression, likelihood, and shrinkage parameters conditional on the fixed reservoir features. The empirical results evaluate this conditional Bayesian quantile-regression analysis under specified reservoir constructions.

The AL and exAL families are used as working likelihoods for conditional quantiles. They provide posterior distributions and conditionally tractable augmented updates. The simulation and application sections report fit recovery, forecast scoring, empirical predictive coverage, monotonicity, and convergence diagnostics to assess the model-based summaries under possible misspecification. Sandwich corrections and generalized-Bayes calibration for composite quantile likelihoods remain important topics for further study.

The multi-quantile construction makes the same distinction. Independent single-level Q-DESN fits followed by isotonic regression or rearrangement produce monotone quantile curves from level-wise regressions. The joint quantile-vector extension shares information across quantile levels through RHS shrinkage on adjacent coefficient differences. This shrinkage can reduce crossings but does not enforce monotonicity. Ordered parameterizations or monotone rearrangement can be used when monotonicity is required.

The single-quantile dynamic simulations assess recovery and held-out forecasting under known

data-generating mechanisms. The joint multi-quantile study assesses quantile-path recovery and crossings under known conditional quantile paths. The GloFAS application studies one retrospective streamflow forecast origin with independent Q-DESN AL-VB fits, blended future weather covariates, monotone reporting, and undercoverage of the nominal 90% interval. The PriceFM comparison uses matched PriceFM forecasts with retrospectively observed lead covariates. Broader rankings of Q-DESN against dynamic quantile or foundation-model alternatives require evaluations across additional data sets, forecast origins, and reservoir specifications.

Several extensions are natural. On the statistical side, future work should study calibrated composite-likelihood scaling, hard or probabilistic non-crossing constraints, richer covariance structures for quantile differences, and sensitivity to reservoir scaling on bounded feature domains. On the computational side, the joint model calls for sparse precision solvers, targeted covariance extraction for VB, and diagnostics comparing VB with MCMC on small joint problems. On the application side, multi-origin operational GloFAS studies with covariates known at the forecast origin, paired PriceFM uncertainty analyses, and explicitly joint PriceFM analyses are needed before the joint quantile-vector prior can be interpreted in an application-level benchmark.

Q-DESN combines fixed nonlinear recurrent features with Bayesian regression for dynamic conditional quantiles. The results demonstrate conditional-quantile estimation under the specified simulation designs and retrospective applications and identify the settings requiring broader validation.

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